

KURUKSHETRA 2.0

HACKFEST 2026

VAANIMUDRA

- Problem ID: NA
 - Problem: Speech to ISL translation with 3D Avatar
 - Theme: Open Innovation
 - Team ID: KH037
 - Team: GoldMiners
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Understanding the Problem



- Deaf and hard-of-hearing users often need to communicate with people who do not understand ISL. Existing tools rarely connect speech, ISL grammar, sign lookup, and avatar playback in one flow.

Problem Analysis

- English and ISL use different grammar.
- Word-for-word translation can lose meaning, tense, questions, and negation.

Who It affects

- Deaf and hard-of-hearing users, families, educators, healthcare staff, and public-service workers.
- 63M+ Deaf & hard-of-hearing persons in India
- 5M+ ISL speakers with limited digital tools



Proposed Solution

Speech or text becomes an ISL avatar

Input: typed text or speech

Output: ISL GLOSS, SiGML, and sequential 3D avatar playback

Innovation & Uniqueness

- Context-aware ISL translation beyond word-to-word conversion
- Hybrid Pipeline with semantic fallback for better sign coverage.
- Real-time 3D avatar playback for accessible communication.

Existing Solutions

Follow fixed rules and struggle with new or different unseen sentences.

Most AI sign tools mainly support ASL or European sign languages, with limited focus on Indian Sign Language (ISL).

Technical approach



Technology stack

React Native + Expo

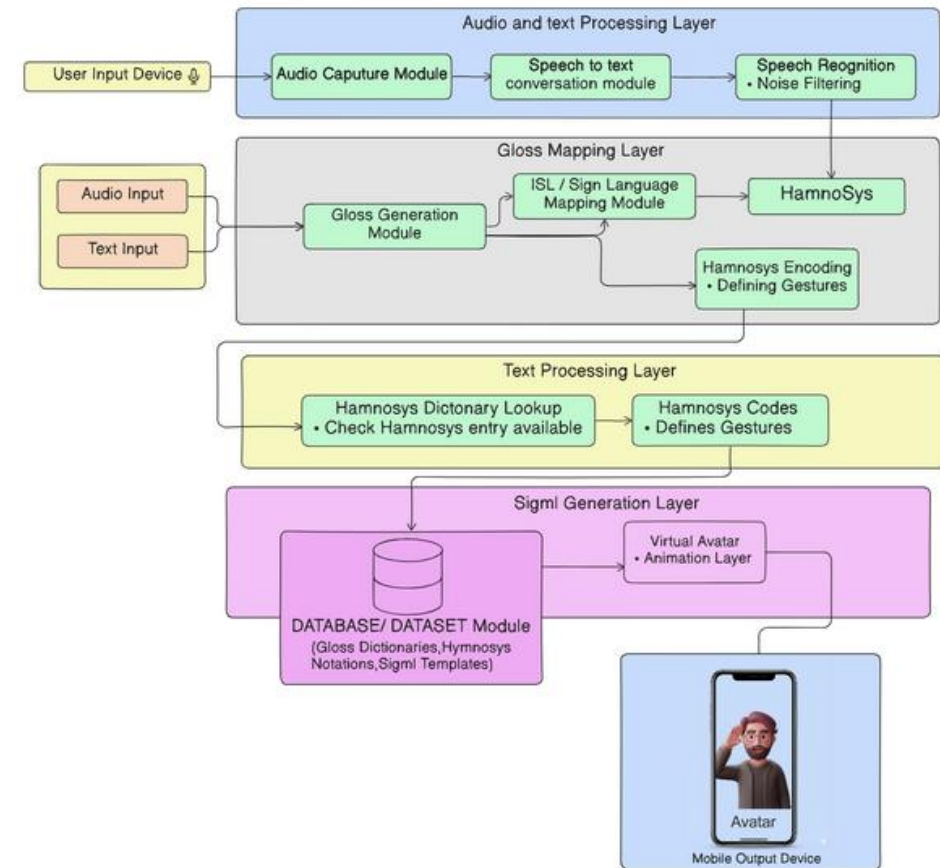
FastAPI + Uvicorn

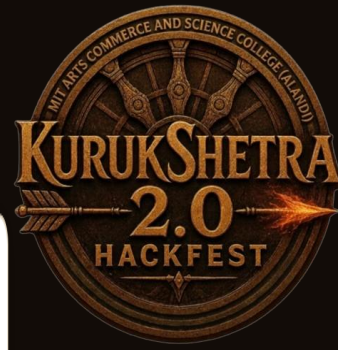
spaCy + Groq API

ChromaDB + SentenceTransformer

Working flow

Speech/text -> NLP -> ISL GLOSS -> sign
lookup -> SiGML -> avatar





Feasibility and impact

Feasible now

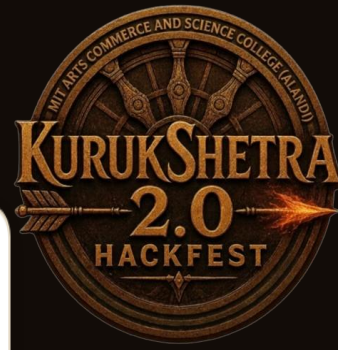
- Built with available React Native, FastAPI, and SiGML technologies.
- Modular architecture supports easy expansion and maintenance.
- Semantic fallback reduces missing-sign limitations.

Impact & Benefits

- Enables communication between ISL users and non-signers.
- Supports education, healthcare, and public-service interactions.
- Provides an affordable and scalable assistive solution.

Current AI and ML

- spaCy extracts linguistic features. A Groq-compatible model creates structured semantic fields.
- Rules generate GLOSS.
- SentenceTransformer and ChromaDB handle semantic retrieval.



Research and references

Research conducted

- ISL grammar, HamNoSys and SiGML,
- CWASA avatar rendering and accessibility design.
- Semantic retrieval, speech recognition, and

Sources

Research paper; ISLRTC: islrtc.nic.in; CWASA: vhg.cmp.uea.ac.uk; SentenceTransformers: [sbert.net](https://www.sbert.net); ChromaDB: docs.trychroma.com

Repository

https://github.com/GoldMiners141/vaa_nimudra/



Build with

Frontend

- React Native, Expo, TypeScript, WebView

Backend

- Python, FastAPI, Uvicorn

Database

- JSON metadata, SiGML, local files

AI/ML

- spaCy, Groq API, SentenceTransformer, ChromaDB

Tools & Services

- CWASA, AsyncStorage, SpeechRecognition

Architecture Principle

- Modular, cross-platform, accessibility-first



THANK YOU